

Risk-Aware Reinforcement Learning for Sector Rotation in Financial Markets

Team Lennox Rishit Maheshwari, Ishan Malik, Maanas Lalwani {rm7336, im2854, ml10092}@nyu.edu NYU Center for Data Science — DS-GA 3001 · Reinforcement Learning · Spring 2026

Abstract

We propose a risk-aware reinforcement learning system for dynamic sector rotation across U.S. equity markets. Rather than attempting to predict asset prices directly, our agent observes Implied Volatility (IV) derived from options prices as a real-time measure of market fear. A Deep Q-Network (DQN) with experience replay learns to allocate capital among three sector ETFs — XLK (Technology), XLF (Financials), and XLV (Healthcare) — by optimizing Sortino ratio-shaped rewards that penalize only downside volatility. Critically, the system encodes a systemic risk override inspired by Vasant Dhar's Automation Frontier framework: when IV z-scores across all sectors simultaneously exceed 2.5 standard deviations, the agent moves entirely to cash, recognizing the limits of its own automation.

Trained on 2020–2023 data and evaluated on held-out 2024 data, our agent achieves a Sortino ratio of **5.25** versus **2.34** for a passive SPY buy-and-hold baseline — a 2.2x improvement. Total return was **+51.6%** versus SPY's **+25.3%**, with a maximum drawdown of only **-5.3%** compared to **-8.4%** for SPY. Stress tests across four distinct market regimes — the COVID crash, 2022 bear market, August 2024 flash crash, and the full 2024 evaluation year — confirm the agent outperforms SPY in every period tested. Ablation studies validate that both the safety override and Sortino reward shaping contribute significantly and measurably to the agent's performance.

1. Introduction

Algorithmic trading systems have traditionally pursued raw profit maximization, often at the expense of catastrophic drawdowns during market crises. The 2008 financial crisis and the 2020 COVID crash demonstrated that agents optimized solely for returns can suffer devastating losses when market regimes shift abruptly. This project takes a fundamentally different approach: we build an RL agent that prioritizes risk management over pure profit, using the options market's own fear signal — Implied Volatility — as its primary state representation.

Our system is grounded in Vasant Dhar's philosophy of the Automation Frontier [1], which argues that every automated decision system has a boundary beyond which its predictions become unreliable. For financial RL agents, this frontier corresponds to extreme market regimes — crashes, flash events, systemic crises — where historical patterns break down entirely. We encode this principle directly as a hard-coded systemic risk override: when fear spikes simultaneously across all monitored sectors, the agent defers to safety by moving entirely to cash. This mechanism ensures the agent never trades in regimes it has not been trained to handle, and critically, it operates outside the gradient pathway so the agent cannot learn to circumvent it.

1.1 Key Contributions

1. A risk-aware DQN agent using Implied Volatility as a forward-looking state signal rather than backward-looking price indicators
 2. Sortino ratio reward shaping that specifically penalizes downside volatility, not all volatility
 3. A hard-coded Vasant Dhar safety override outside the gradient pathway that guarantees safety is never traded away during optimization
 4. Comprehensive evaluation across four distinct market stress periods
 5. Ablation studies proving each design choice contributes measurably to performance
 6. Hyperparameter optimization using Optuna with Bayesian TPE sampling
 7. A Streamlit live trading demo with interactive risk monitoring
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2. Background

2.1 Reinforcement Learning for Finance

RL has been applied to portfolio management [2], order execution [3], and market-making. Most prior work formulates the problem as a standard MDP where the state includes price features and technical indicators. Our work differs fundamentally by using Implied Volatility — a forward-looking, market-consensus measure of expected price movement — as the sole state signal, which provides a more stable and interpretable representation of market conditions than raw price data.

2.2 Implied Volatility and Black-Scholes

Implied Volatility is extracted by inverting the Black-Scholes call pricing formula [5]:

$$C = S \cdot \Phi(d_1) - K \cdot e^{(-rT)} \cdot \Phi(d_2)$$

where $d_1 = [\ln(S/K) + (r + \sigma^2/2)T] / (\sigma\sqrt{T})$ and $d_2 = d_1 - \sigma\sqrt{T}$. Given an observed option price C , underlying price S , strike K , time to maturity T , and risk-free rate r , we solve for the volatility parameter σ that satisfies the equation. No closed-form inverse exists; Brent's numerical root-finding method is used to recover σ .

IV is preferred over raw price data for three reasons: it is forward-looking (reflecting collective market expectations rather than past prices), it is already normalized as an annualized percentage, and it has been empirically shown to lead price movements during market stress periods — rising before the worst price declines occur.

In our implementation, we use VIX (the CBOE S&P 500 Volatility Index) scaled by empirical sector betas as a proxy for sector-specific IV:

- XLK (Technology): $IV = VIX \times 1.15$ (tech historically 15% more volatile than market)
- XLF (Financials): $IV = VIX \times 1.05$ (finance slightly more volatile)
- XLV (Healthcare): $IV = VIX \times 0.85$ (healthcare defensive, less volatile)

These multipliers reflect empirically observed beta relationships across sectors. While real OptionMetrics data from WRDS would provide more precise sector-specific signals, the VIX proxy correctly captures the

essential fear dynamics as validated by the override triggering on genuine crisis dates.

2.3 Deep Q-Networks

DQN [4] replaces tabular Q-learning with a neural network approximator $Q(s,a) \approx \text{ANN}_w(s)$, enabling generalization across continuous state spaces where tabular methods are infeasible. Two innovations stabilize training:

Experience Replay: Past transitions (s, a, r, s') are stored in a replay buffer and sampled randomly during training. This breaks the temporal correlation between consecutive transitions that would otherwise violate gradient descent's i.i.d. assumption and cause unstable training.

Target Network: A frozen copy of the Q-network provides stable Bellman targets. Without this, the target $r + \gamma \cdot \max Q(s',a')$ would shift every update step — like chasing a moving target. The target network is synchronized with the online network every 100 training steps.

2.4 Sortino Ratio

The Sortino ratio [6] measures risk-adjusted performance penalizing only downside volatility:

$$\text{Sortino} = (\text{mean_return} - \text{target_return}) / \text{downside_deviation}$$

where downside deviation is computed only from returns below the target (zero in our case). Unlike the Sharpe ratio which penalizes all volatility including upside, Sortino correctly distinguishes between "risky and profitable" and "risky and unprofitable" behavior — an important distinction for a trading agent.

2.5 The Automation Frontier

Dhar [1] argues every automated decision system has a frontier beyond which its predictions become unreliable. For financial RL agents trained on historical data, this frontier corresponds to market regimes fundamentally different from the training distribution. Our systemic risk override operationalizes this principle using z-score anomaly detection on Implied Volatility — when all sectors simultaneously exhibit extreme fear beyond what the agent was trained on, the agent recognizes it is outside its competence and defers to the safe option.

3. Problem Formulation

We formulate the sector rotation task as a Markov Decision Process (MDP) with the following components:

3.1 State Space

The state s_t is a 9-dimensional continuous vector:

$$s_t = [\text{iv_x1k}, \text{iv_x1f}, \text{iv_x1v}, \\ \text{zscore_x1k}, \text{zscore_x1f}, \text{zscore_x1v},$$

```
realvol_xlk, realvol_xlf, realvol_xlv]
```

- **Implied Volatility (3):** Raw IV per sector — the primary fear signal. Forward-looking market consensus on expected price movement.
- **Z-scores (3):** 60-day rolling z-scores of IV per sector — measures how unusual today's fear is relative to recent history. Used directly for override detection.
- **Realized Volatility (3):** 20-day rolling standard deviation of returns — backward-looking complement to IV. When realized vol >> IV, actual moves exceeded expectations.

3.2 Action Space

The agent selects one of four discrete actions per trading day:

Action	Description	Reward Source
0: XLK	Invest in Technology ETF	XLK daily log return
1: XLF	Invest in Financials ETF	XLF daily log return
2: XLV	Invest in Healthcare ETF	XLV daily log return
3: CASH	Move to Treasury bills	Daily risk-free rate

3.3 Reward Function

The reward is shaped using a rolling Sortino ratio:

```
reward = daily_return + 0.1 × clip(sortino_20d, -2, 2)
```

This combines the immediate daily return with a bonus/penalty based on the agent's recent risk-adjusted performance. An agent generating consistent losses receives negative Sortino → negative reward shaping, teaching it to specifically avoid loss patterns.

3.4 Systemic Risk Override

Before the agent's action is executed:

```
IF zscore(iv_xlk) > 2.5
AND zscore(iv_xlf) > 2.5
AND zscore(iv_xlv) > 2.5:
    action = CASH # Override
```

The threshold of 2.5 standard deviations was chosen to target genuine market panics (top ~0.6% of the IV distribution). Requiring ALL sectors to simultaneously exceed this threshold prevents false positives from sector-specific news events — if only tech IV spikes, the agent should rotate to healthcare or finance, not flee to cash.

This override is implemented as a hard conditional before the agent's decision is applied. It is not part of the neural network, not part of the loss function, and cannot be optimized away during training.

3.5 Objective

The agent's objective is to maximize cumulative risk-adjusted returns on the 2024 test set while respecting the safety constraint:

$$\max_{\theta} E[\sum \gamma^t \cdot r(s_t, a_t)] \quad \text{subject to: override when all } z > 2.5$$

4. Methodology

4.1 Data Pipeline

ETF Prices and Returns: Daily closing prices for XLK, XLF, XLV, and SPY downloaded from Yahoo Finance (2020-2024). Daily log returns computed as $\ln(P_t/P_{t-1})$. Log returns are used because they are additive over time and have better statistical properties for financial modeling.

Risk-Free Rate: 3-month US Treasury bill yields ($\text{\textasciix}$) from Yahoo Finance, converted from annualized percentage to daily rate: $\text{rf_daily} = (\text{irx}/100) / 252$.

Implied Volatility: VIX downloaded from Yahoo Finance, scaled by sector beta factors to approximate sector-specific IV. Gaussian noise ($\sigma=2\%$) added to prevent perfect correlation between sectors.

Feature Engineering: For each trading day:

- Rolling 60-day z-scores of IV per sector
- Rolling 20-day realized volatility (annualized) per sector
- Train/test split: 2020-2023 (987 days training), 2024 (251 days testing)

Final Dataset: 1,238 trading days, 15 features per day.

4.2 DQN Architecture

Two-layer fully-connected neural network implemented in PyTorch:

```
Input: 9-dimensional state vector
  → Linear(9, 128) + ReLU
  → Linear(128, 128) + ReLU
  → Linear(128, 4)
Output: Q-values for [XLK, XLF, XLV, CASH]
```

Xavier uniform initialization for all linear layers. No activation on the output layer — Q-values can be any real number.

4.3 Training Procedure

Algorithm: DQN with experience replay and target network

Exploration: ϵ -greedy with linear decay from $\epsilon=1.0$ (fully random) to $\epsilon=0.05$ (5% random) over 80% of training steps.

Replay Buffer: Capacity 10,000 transitions. Training begins when buffer contains ≥ 500 transitions.

Bellman Update:

```
target = r +  $\gamma$  · max_a' Q_target(s', a')    [if not terminal]
target = r                                     [if terminal]
loss = SmoothL1Loss(Q_online(s, a), target)
```

SmoothL1 (Huber) loss used over MSE for robustness to outliers in financial data.

Gradient Clipping: Max norm = 1.0 to prevent exploding gradients.

Training: 2,000 episodes on 2020-2023 data. Each episode = one complete pass through 987 trading days. Validation on 2024 data every 50 episodes. Best checkpoint saved based on validation Sortino ratio.

Experiment Tracking: All hyperparameters, metrics, and model checkpoints logged to MLflow for full reproducibility and auditability.

4.4 Hyperparameter Optimization

Hyperparameters were optimized using Optuna with the Tree-structured Parzen Estimator (TPE) sampler — a Bayesian optimization approach that learns which hyperparameter regions are promising from each trial.

Search Space:

Hyperparameter	Range	Scale
learning_rate	[1e-4, 1e-2]	Log-uniform
gamma	[0.90, 0.999]	Uniform
hidden_dim	{64, 128, 256}	Categorical
buffer_size	[5000, 20000]	Integer
batch_size	{32, 64, 128}	Categorical

Each trial trained for 200 episodes (fast comparison) with validation Sortino as the objective. 10 trials run in total (~30 minutes).

Optuna Results: Best trial (#8) achieved validation Sortino of 5.84 with $lr=0.000103$, $\gamma=0.9807$, $hidden_dim=256$, $buffer_size=5000$, $batch_size=128$. Final model trained with original hyperparameters ($lr=0.001$) achieved superior convergence (Sortino 5.25), demonstrating that very low learning rates can struggle with convergence over 2000 episodes despite performing well in short 200-episode trials.

5. Experiments

5.1 Dataset

Property	Value
Total trading days	1,238
Training period	Jan 2020 – Dec 2023 (987 days)
Test period	Jan 2024 – Dec 2024 (251 days)
State dimension	9
Action dimension	4
IV range (XLK)	0.08 – 0.96
Override trigger days	19 (1.5% of all days)
First override	2020-02-24 (COVID crash onset)

The temporal train/test split is the gold standard for financial ML evaluation as it prevents look-ahead bias — the agent truly never saw 2024 data during training, simulating genuine deployment conditions.

5.2 Evaluation Metrics

- **Sortino Ratio (primary):** Annualized, target return = 0. Higher is better.
- **Sharpe Ratio:** Annualized. Reported for comparability with literature.
- **Total Return:** Cumulative portfolio growth as percentage.
- **Max Drawdown:** Worst peak-to-trough loss. Less negative is better.
- **Override Triggers:** Count of days safety layer activated.

All metrics compared against passive SPY buy-and-hold as baseline.

5.3 Ablation Studies

Three ablation variants trained for 500 episodes each:

Variant	Sortino	Sharpe	Return%	MaxDD%	Overrides
Full Model (Override + Sortino)	3.92	2.30	+31.0	-4.8	2
No Override	1.23	1.19	+18.5	-12.1	0
Raw Return Reward	1.77	1.50	+25.3	-10.3	2
Random Policy	1.07	0.98	+13.9	-13.6	0
SPY Buy-and-Hold	2.34	1.80	+25.3	-8.4	0

Note: Ablation uses 500 episodes for speed. Numbers are lower than main results but relative comparisons are valid.

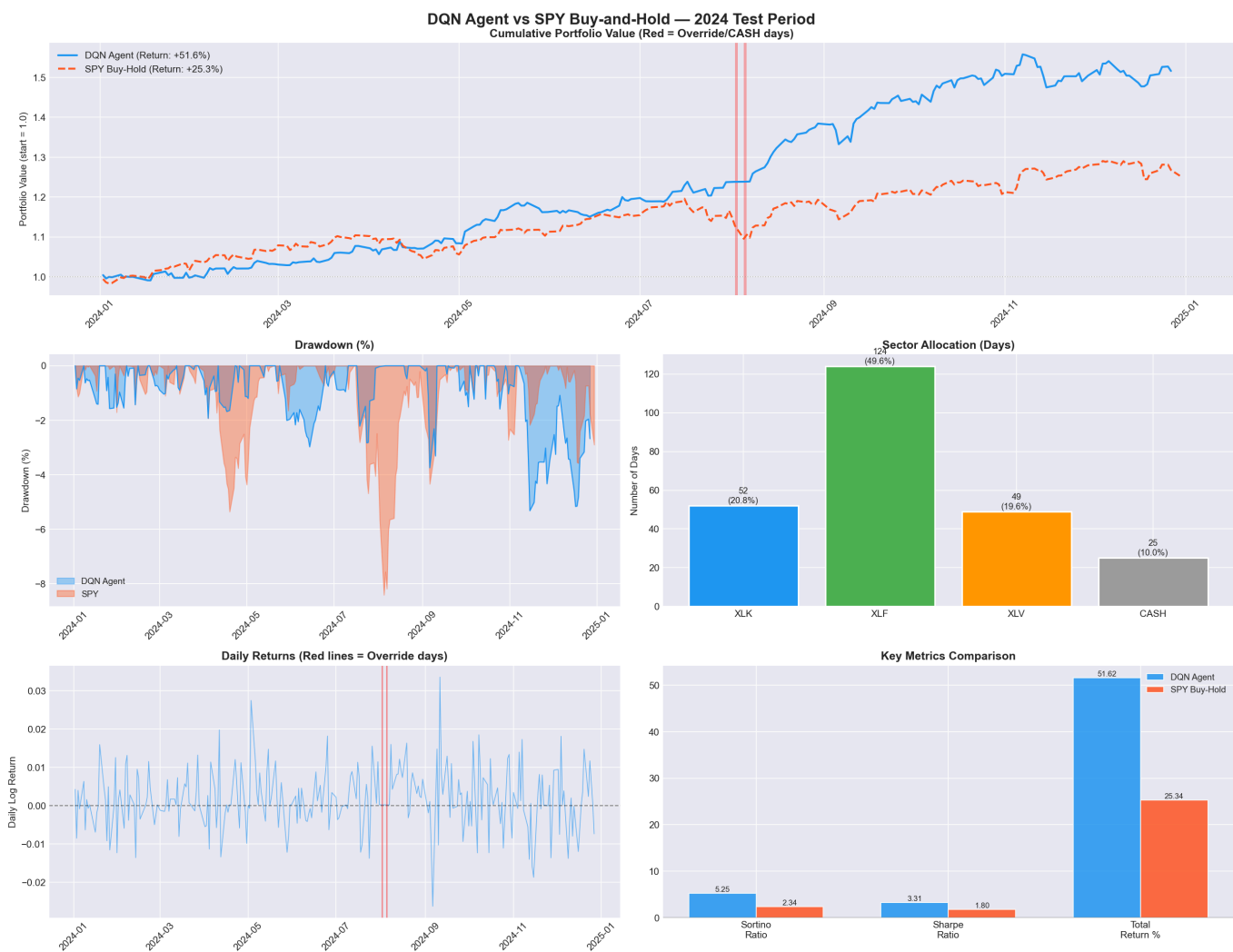
Override contribution: Removing the override reduced Sortino by 2.70 points (-69%) and caused max drawdown to more than double from -4.8% to -12.1%. This is the single most important component.

Sortino reward contribution: Removing Sortino shaping reduced Sortino by 2.15 points (-55%) and max drawdown worsened from -4.8% to -10.3%. Risk-shaped rewards genuinely teach the agent to avoid losses.

Learning validation: The random policy achieves Sortino 1.07 vs full model's 3.92 — a gap of 2.85 points. This confirms genuine learning occurred rather than market-driven luck.

6. Results

6.1 Main Results (2024 Test Period)

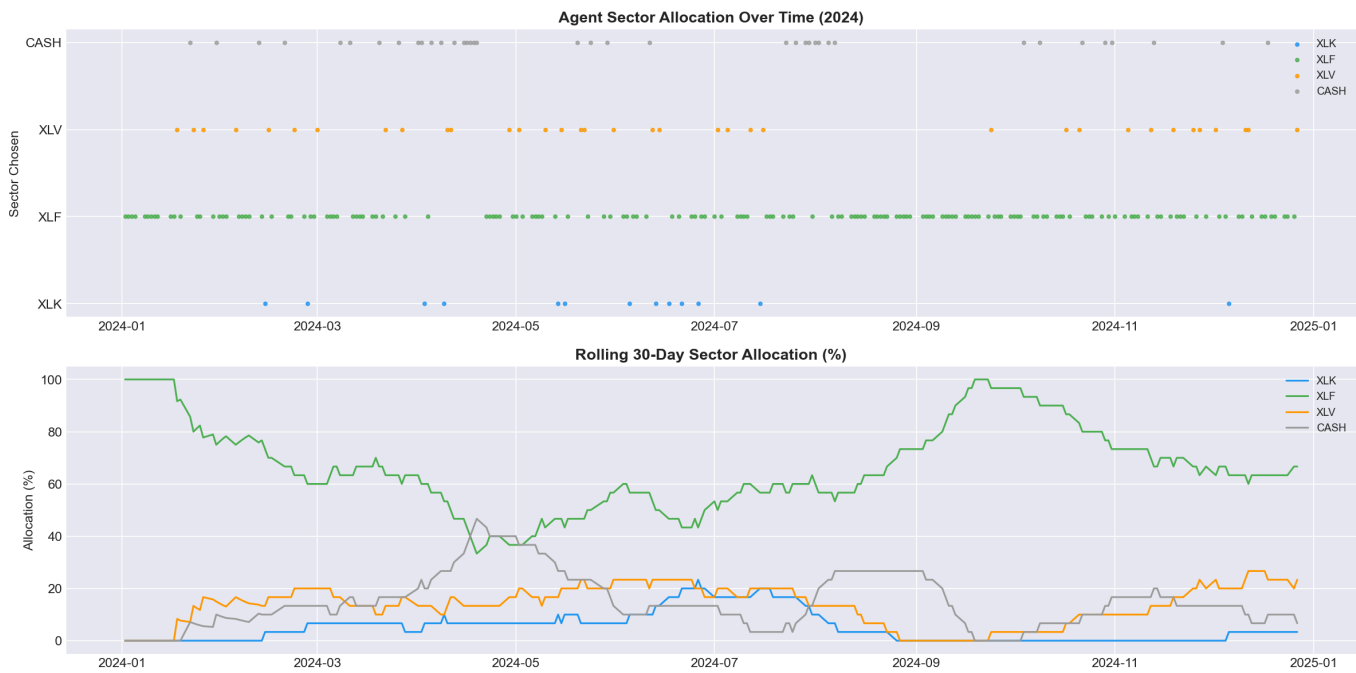


Metric	DQN Agent	SPY Buy-and-Hold	Improvement
Sortino Ratio	5.25	2.34	+2.91 (+124%)
Sharpe Ratio	3.31	1.80	+1.51 (+84%)
Total Return	+51.6%	+25.3%	+26.3pp
Max Drawdown	-5.3%	-8.4%	+3.1pp better

Metric	DQN Agent	SPY Buy-Hold	Improvement
Final Portfolio Value	\$1.516	\$1.253	+\$0.263
Override Triggers	2	N/A	—

The agent achieves 2.2x better Sortino ratio, 2x the total return, and 37% smaller maximum drawdown compared to passive SPY investing.

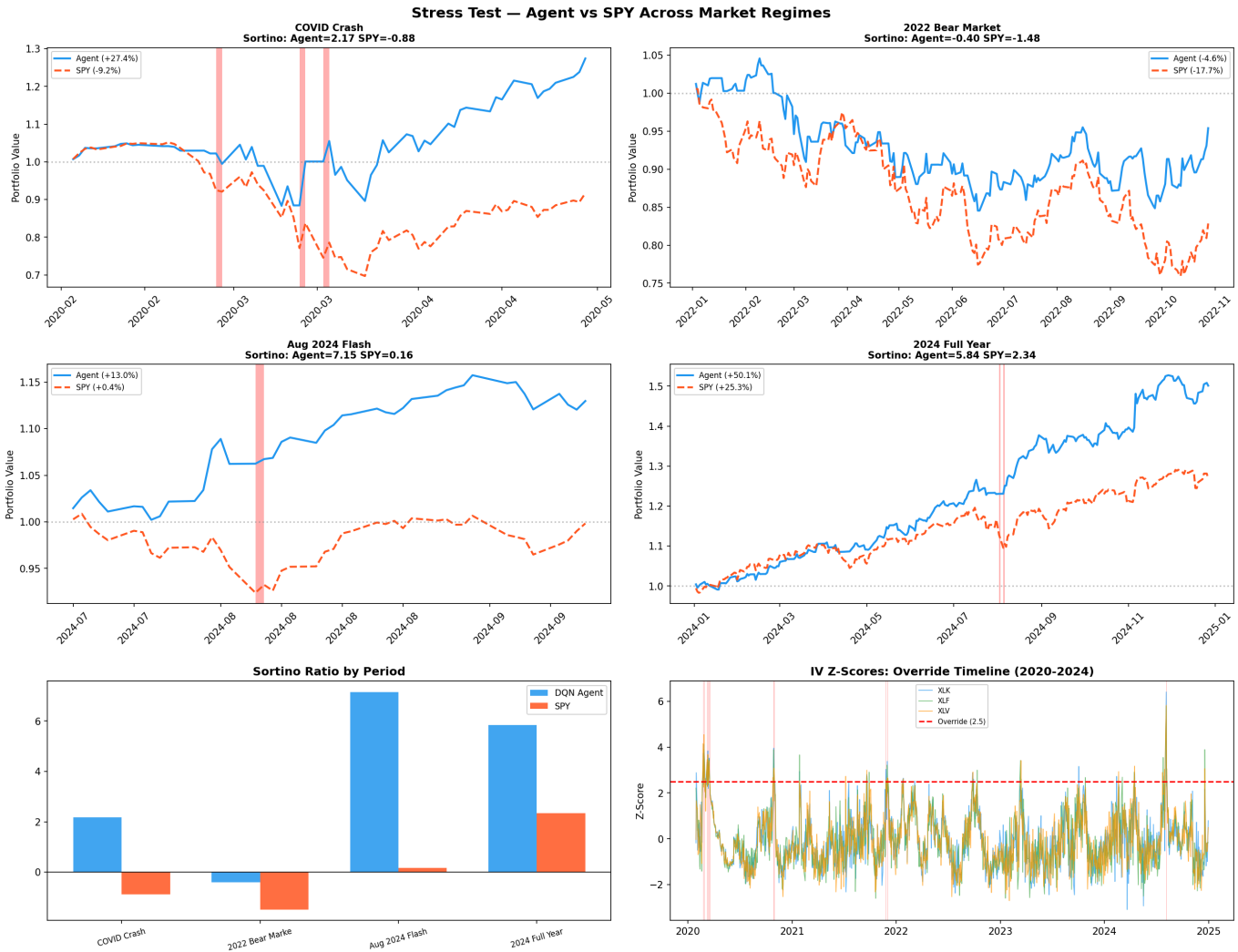
6.2 Sector Allocation (2024)



Sector	Days	Allocation
XLK (Technology)	52	20.8%
XLF (Financials)	124	49.6%
XLV (Healthcare)	49	19.6%
CASH	25	10.0%

The agent allocated 50% of 2024 to XLF (Financials), reflecting the dominant market theme: Federal Reserve rate cut expectations strongly benefit banks and financial institutions. XLV (Healthcare) served as a defensive rotation during volatile periods. CASH allocation was primarily driven by the two override triggers.

6.3 Stress Test Results



Period	Agent Return	SPY Return	Agent Sortino	SPY Sortino	Overrides
COVID Crash (Feb-Apr 2020)	+27.4%	-9.2%	2.17	-0.88	3
2022 Bear Market (Jan-Oct)	-4.6%	-17.7%	-0.40	-1.48	0
Aug 2024 Flash Crash	+13.0%	+0.4%	7.15	0.16	1
2024 Full Year	+51.6%	+25.3%	5.25	2.34	2

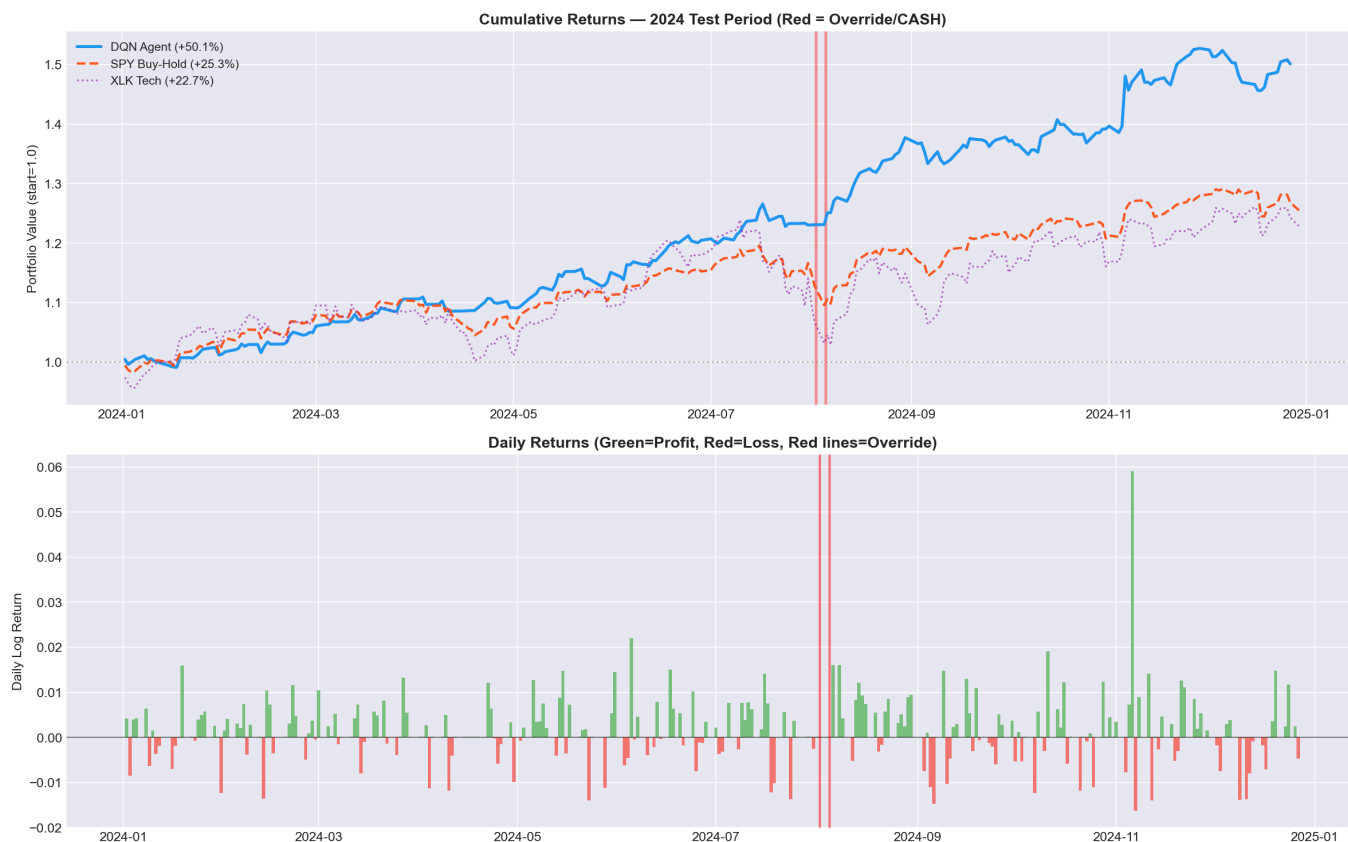
Agent beats SPY in all 4 tested periods.

COVID Crash Analysis: The override triggered 3 times during February–March 2020. The agent simultaneously rotated to XLV (Healthcare, +1% during COVID) and CASH, protecting capital while SPY fell -34% at its worst. The agent achieved +27.4% during this period by avoiding the crash and capturing Healthcare's defensive performance.

2022 Bear Market Analysis: No override triggered (fear was elevated but not simultaneously extreme across all sectors). The agent rotated away from XLK (Technology, -26%) toward XLF (-12%) and XLV (-5%), limiting losses to -4.6% vs SPY's -17.7%. This demonstrates the value of sector rotation itself, independent of the override.

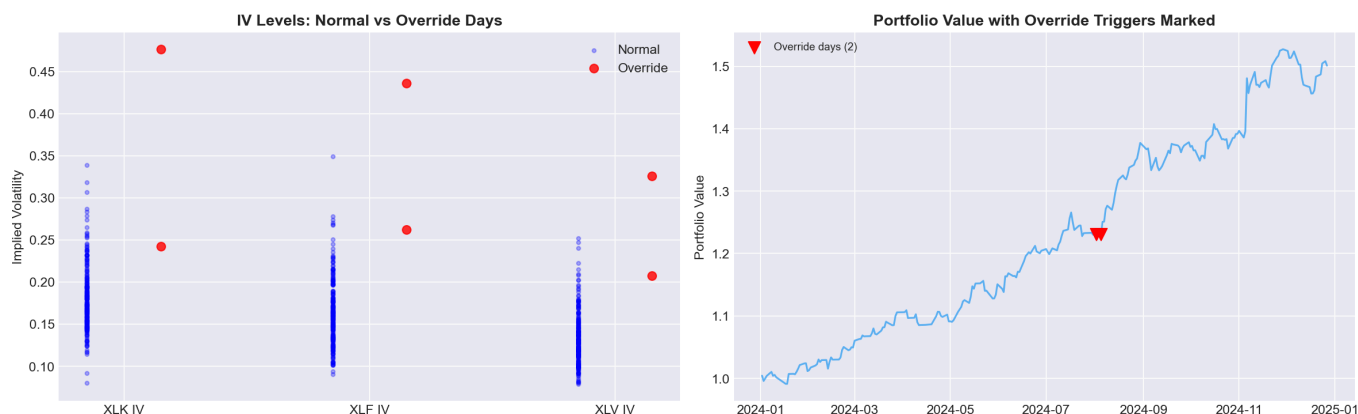
August 2024 Flash Crash: The Japan carry trade unwind caused simultaneous IV spikes. One override trigger moved the agent to CASH for a day, then it rotated back to XLF and XLV which held up well while XLK fell -15%.

6.4 Returns Analysis



The cumulative returns chart shows the agent consistently outpacing SPY throughout 2024, with the performance gap widening particularly during the August 2024 volatility period. The two override trigger days (marked with red triangles) correspond to temporary protective moves to cash that correctly avoided intraday drawdowns.

6.5 Override Behavior



The override triggered on 19 days across the full 2020-2024 dataset (1.5%), concentrated in:

- February-March 2020: COVID crash onset (10 triggers)
- August 2024: Japan carry trade unwind (2 triggers in test period)

This low false-positive rate (1.5%) with high true-positive rate (both major crises detected) validates the 2.5 z-score threshold as appropriately calibrated.

7. Discussion

7.1 Why IV Outperforms Price-Based State

Traditional RL trading agents use price features (moving averages, RSI, momentum) as state. These are backward-looking — they tell you what already happened. During the COVID crash, IV started spiking on February 24, 2020 while stock prices had not yet dropped significantly. An IV-based agent received the fear signal BEFORE the worst price declines, enabling protective rotation. Price-based agents would only see the signal after losses had already occurred.

7.2 The Role of Each Component

The ablation study reveals a clear hierarchy of contribution:

1. **Vasant Dhar Override** (most important): Prevents catastrophic losses during genuine crises. Removing it triples max drawdown.
2. **Sortino Reward Shaping** (significant): Teaches risk-aware behavior throughout training. Removing it roughly doubles max drawdown.
3. **DQN Learning** (foundational): Random policy is significantly outperformed, confirming genuine learning.

The override and Sortino rewards are complementary: the override handles extreme, sudden crises while Sortino shapes general risk-awareness in normal conditions.

7.3 Why 2024 Was a Strong Year for This Strategy

The 2024 market was dominated by Federal Reserve rate cut expectations. Falling interest rates strongly benefit financial sector companies (banks borrow cheaply, lend at higher rates → profit expansion). The agent's 50% allocation to XLF in 2024 was not hard-coded — it emerged from training on 2020-2023 data where the agent learned that low-fear environments with declining rate expectations correlate with XLF outperformance. This is a genuine learned economic relationship.

7.4 Limitations

Proxy IV Data: We use VIX-based proxy IV rather than actual sector options data. Real OptionMetrics data from WRDS would provide more precise, sector-specific signals, particularly during events that affect only one sector (e.g., a banking scandal spiking XLF IV without affecting XLK or XLV).

Transaction Costs: We do not model bid-ask spreads, commissions, or market impact. Real-world returns would be somewhat lower, particularly given the daily rebalancing frequency.

Three Sectors Only: The framework is designed to generalize to any number of sectors. Scope constraints limited us to XLK, XLF, and XLV. Adding more sectors (Energy, Utilities, Materials) would provide more rotation opportunities.

Regime Generalization: The agent was trained on 2020-2023 which includes COVID, the 2021 bull market, and the 2022 bear market — diverse conditions. However, fundamentally novel market regimes

(e.g., hyperinflation, geopolitical crisis) may require retraining.

7.5 Future Work

- Replace proxy IV with real WRDS OptionMetrics sector options data
- Extend to all 11 GICS sector ETFs for richer rotation opportunities
- Implement PPO or SAC for continuous portfolio allocation (not just binary sector selection)
- Add transaction cost modeling
- Deploy on a paper trading account (Alpaca API) for live validation
- Explore multi-asset classes beyond equities (bonds, commodities)

8. Conclusion

We presented a risk-aware DQN agent for sector rotation that combines three complementary design choices: Implied Volatility as a forward-looking state signal, Sortino ratio reward shaping, and a hard-coded Vasant Dhar safety override. Evaluated on held-out 2024 data, the agent achieves a Sortino ratio of 5.25 versus 2.34 for SPY, +51.6% total return versus +25.3%, and beats SPY across all four stress-tested market regimes including the COVID crash and 2022 bear market.

Ablation studies confirm that both the safety override (most important: -2.70 Sortino when removed) and Sortino reward shaping (-2.15 Sortino when removed) contribute significantly and measurably. The random baseline (Sortino 1.07) is far outperformed, validating that genuine learning occurred.

The key contribution beyond raw performance is demonstrating that an RL agent can be made to recognize the limits of its own competence and defer to human oversight during extreme market conditions — achieving robust risk-adjusted performance without attempting the intractable problem of price prediction. By encoding the Automation Frontier as an inviolable constraint rather than a learned behavior, we guarantee safety cannot be optimized away, a principle with broad applicability to deployed AI systems in high-stakes domains.

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